Berkeley Out-of-Order Machine (BOOM) Setup

Created by NIGIL MOHRA

1. Setting-up Chipyard on Ubuntu

```
# Updates
sudo apt update
sudo apt install -y build-essential
sudo apt install -y git
sudo apt install -y curl
sudo apt install -y wget
sudo apt install -y autoconf
sudo apt install -y automake
sudo apt install -y libtool
sudo apt install -y pkg-config
sudo apt install -y bc
sudo apt install -y zlib1g-dev
sudo apt install -y libglib2.0-dev
sudo apt install -y libpixman-1-dev
sudo apt install -y device-tree-compiler
sudo apt install -y ninja-build
sudo apt install -y openssh-client
```

```
# Installation of Miniconda Environment
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
bash Miniconda3-latest-Linux-x86_64.sh
# Close and Reopen Terminal

# Check
conda config --show auto_activate

# Agree to Terms and Conditions
conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/main
conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/r

# Restart the Terminal
conda config --set auto_activate_base false

# Clone Chipyard
git clone https://github.com/ucb-bar/chipyard.git
mv chipyard Chipyard
cd Chipyard

# Updating the YAML Configuration File
nano conda-reqs/chipyard-base.yaml
```

As before: change `- sysroot_linux-64=2.43` to `- sysroot_linux-64`, and add `- python=3.10` under `dependencies`.

```
# Disable SPEC2026 Cloning
cd ~/Chipyard
git config -f .gitmodules submodule.software/spec2026.update none
git submodule sync

# Cloning Submodules
./scripts/init-submodules-no-riscv-tools.sh
./build-setup.sh riscv-tools -ud --use-lean-conda --skip-submodules

# Cleaning the Conda Environment
rm -rf .conda-lock-env
```

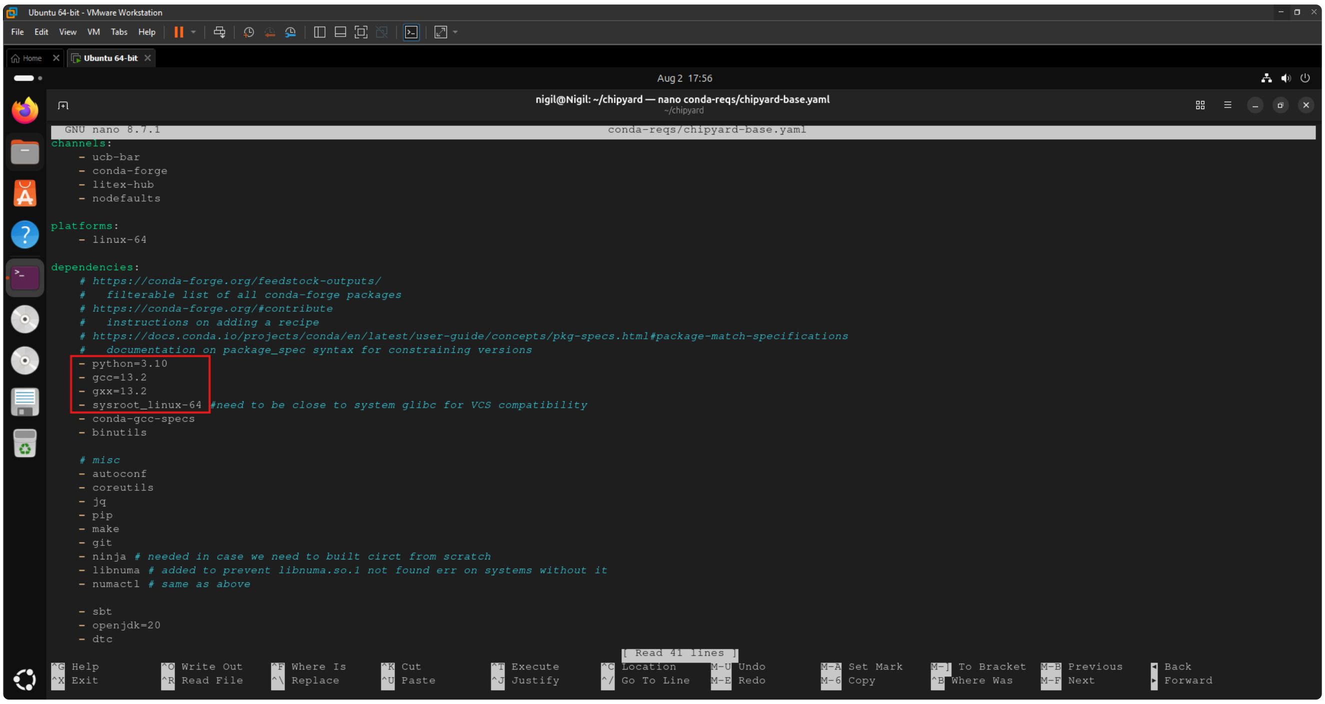

Figure 1. Updating the YAML Configuration File

```
# Install CIRCT Submodule
cd ~/Chipyard

 # Invoke the Conda Environment
source ~/Chipyard/env.sh
git submodule update --init tools/install-circt

# Install
./tools/install-circt/bin/download-release-or-nightly-circt.sh \
    -f circt-full-static-linux-x64.tar.gz \
    -i "$RISCV" \
    -v version-file \
    -x conda-reqs/circt.json \
    -g unused

# Check
which firtool
firtool --version
```

The setup of the required Chipyard submodules for using the BOOM processor is now complete. The next step is to build the BOOM processor by executing the following commands.

---

2. Smoke Test on RV64-BOOM

```
# Configuring BOOM
cd ~/Chipyard
source ~/Chipyard/env.sh
cd sims/verilator
make CONFIG=SmallBoomV3Config debug

# Check
echo $?
ls -lh ~/Chipyard/sims/verilator/simulator-chipyard.harness-SmallBoomV3Config-debug
```

```
# Create Directory
```

```bash
mkdir -p ~/Chipyard/software/AESTest

# Clone Repository
cd ~/Chipyard/software/AESTest
git clone https://github.com/kokke/tiny-AES-C.git .

# Invoke Chipyard Conda Environment
source ~/Chipyard/env.sh
riscv64-unknown-elf-gcc -static -specs=htif_nano.specs -O3 aes.c test.c -o tiny_aes.riscv

# Run
cd ~/Chipyard/sims/verilator
./simulator-chipyard.harness-SmallBoomV3Config-debug \
    +permissive +max-cycles=10000000 +permissive-off \
    ~/Chipyard/software/AESTest/tiny_aes.riscv
```